

Propositional Logic

Outline

I. Syntax

II. Semantics and truth table

III. Knowledge base for the Wumpus world

IV. Inference by model checking

V. Inference by rules

I. Syntax of Propositional Logic

An *atomic sentence* is a single proposition symbol.

standing for a proposition that has to
be either true or false but not both.

$P, Q, R, W_{1,3}, FacingEast$

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Two special proposition symbols:

True : always-true proposition

False: always-false proposition

Complex Sentences

A *complex sentence* is constructed from simpler sentences, using parentheses and *logical connectives* (5 in total).

- $\neg$ (not).
 - ◆ $\neg P$ is the *negation* of P .
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- $\wedge$ (and).
 - ◆ $W_{1,3} \wedge P_{3,1}$ is a *conjunction* whose parts $W_{1,3}$ and $P_{3,1}$ are *conjuncts*.
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There is a pit in [3,1].
- $\vee$ (or).
 - ♦ $(W_{1,3} \wedge P_{3,1}) \vee W_{2,2}$ is a *disjunction* whose parts $(W_{1,3} \wedge P_{3,1})$ and $W_{2,2}$ are *disjuncts*.

More Logical Connectives

- $\Rightarrow$ (implies).

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- $\Leftrightarrow$ (if and only if).

◆ $W_{1,3} \Leftrightarrow W_{2,2}$ is a *biconditional*.

Grammar of Propositional Logic



John Backus (IBM)

National Medal of Science (1975)

ACM Turing Award (1977)

Backus-Naur form (BNF):

$$\textit{Sentence} \rightarrow \textit{AtomicSentence} \mid \textit{ComplexSentence}$$
$$\textit{AtomicSentence} \rightarrow \textit{True} \mid \textit{False} \mid P \mid Q \mid R \mid \dots$$
$$\textit{ComplexSentence} \rightarrow (\textit{Sentence})$$
$$\mid \neg \textit{Sentence}$$
$$\mid \textit{Sentence} \wedge \textit{Sentence}$$
$$\mid \textit{Sentence} \vee \textit{Sentence}$$
$$\mid \textit{Sentence} \Rightarrow \textit{Sentence}$$
$$\mid \textit{Sentence} \Leftrightarrow \textit{Sentence}$$

OPERATOR PRECEDENCE : $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$



Peter Naur (U. Copenhagen)

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OPERATOR PRECEDENCE : $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$



Peter Naur (U. Copenhagen)
ACM Turing Award (2005)

$\neg A \vee B \wedge C \Rightarrow D$ is equivalent to
 $((\neg A) \vee (B \wedge C)) \Rightarrow D$

II. Semantics of Propositional Logic

A *model* fixes the truth value (*true* or *false*) for every proposition symbols.

$$m_1 = \{P_{1,2} = \text{false}, P_{2,2} = \text{false}, P_{3,1} = \text{true}\}$$

$$m_2 = \{P_{1,2} = \text{true}, P_{2,2} = \text{false}, P_{3,1} = \text{true}\}$$

Semantics defines the rules for determining the truth of a sentence w.r.t. any model.

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Semantics defines the rules for determining the truth of a sentence w.r.t. any model.

The truth value of any sentence can be computed once we know

- how to evaluate the truth of atomic sentences;
- how to compute the truth of sentences formed with each of the five connectives.

Determining the Truth Value

Atomic sentences:

- ♦ *true* is true in every model.
- ♦ *false* is false in every model.
- ♦ The truth value of every other proposition symbol must be specified in a model m .

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Complex sentences in the model m :

- ♦ $\neg P$ is true iff P is false.
- ♦ $P \wedge Q$ is true iff P and Q are true.
- ♦ $P \vee Q$ is true iff either P or Q is true.
- ♦ $P \Rightarrow Q$ is true unless P is true and Q is false.
- ♦ $P \Leftrightarrow Q$ is true iff P and Q are both true or both false.

Truth Table

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
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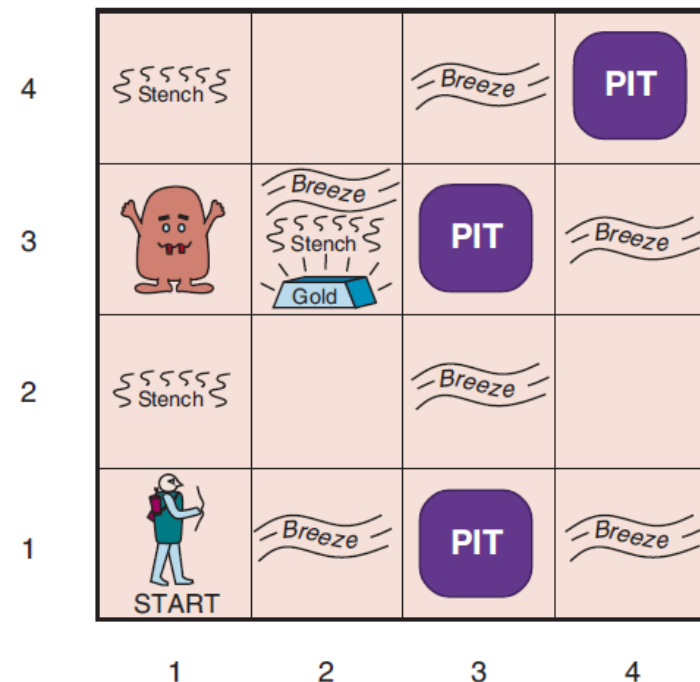
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III. Knowledge Base for the Wumpus World

Proposition symbols

- $P_{x,y}$ is true if there is a pit in $[x, y]$.
- $W_{x,y}$ is true if there is a wumpus in $[x, y]$, dead or alive.
- $B_{x,y}$ is true if the agent perceives a breeze in $[x, y]$.
- $S_{x,y}$ is true if the agent perceives a stench in $[x, y]$.

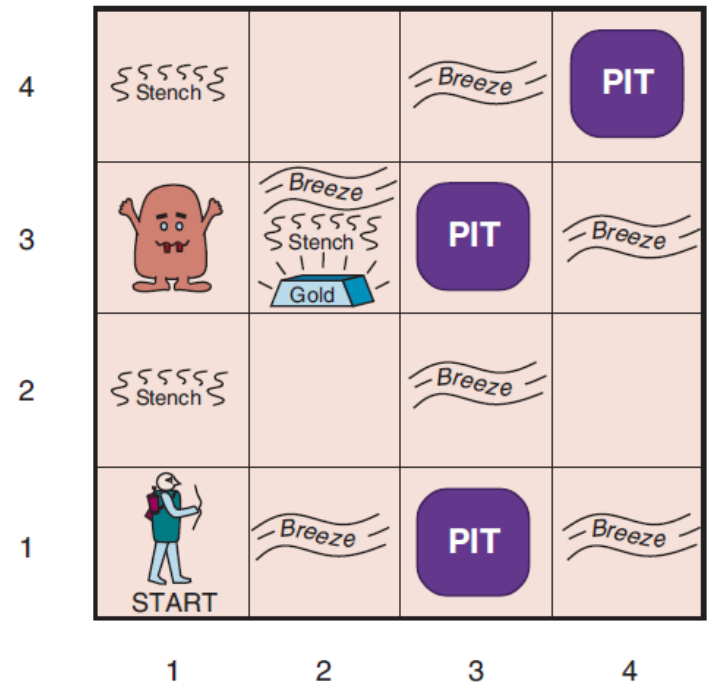


Knowledge Base (cont'd)

◆ General knowledge (partial – only for relevant squares):

- There is no pit in [1,1].

$$R_1: \neg P_{1,1}$$



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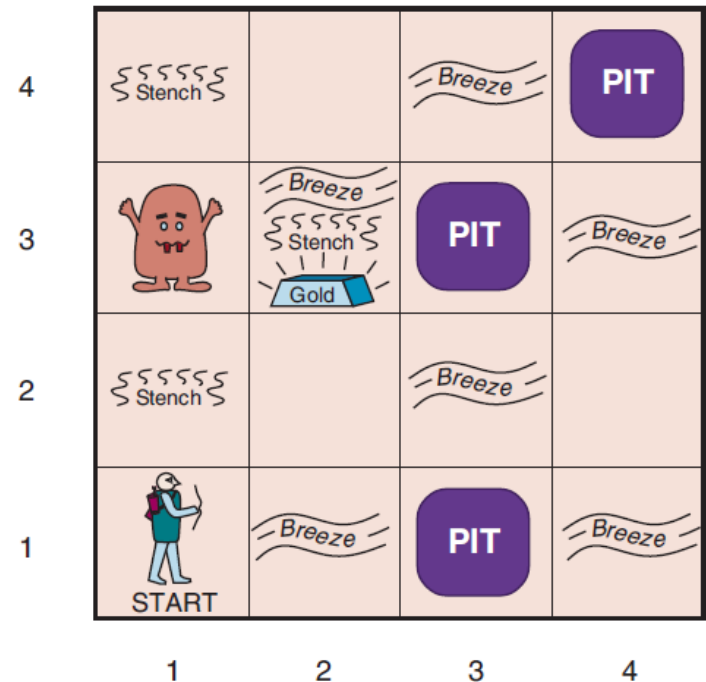
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- A square is breezy if and only if a neighboring square has a pit.

$$R_2: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

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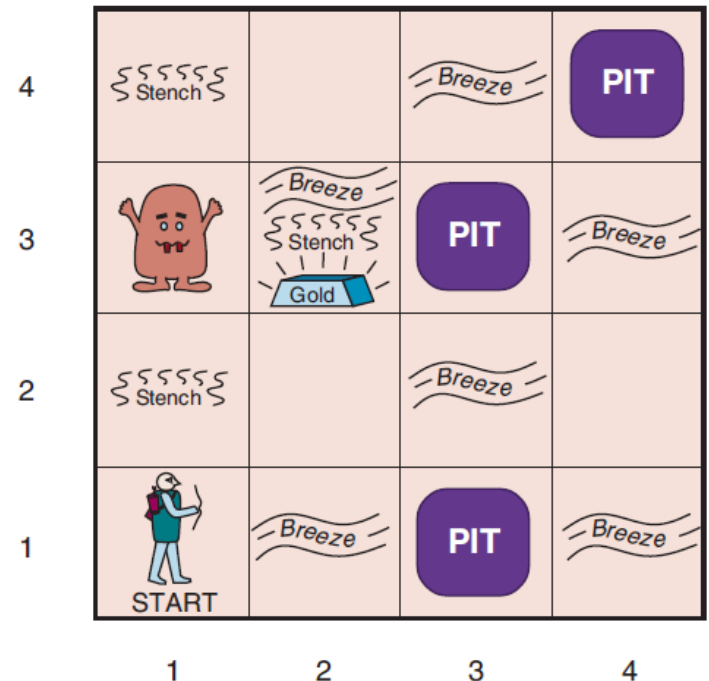
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◆ Percepts for the first two squares:

$$R_4: \neg B_{1,1}$$

$$R_5: B_{2,1}$$



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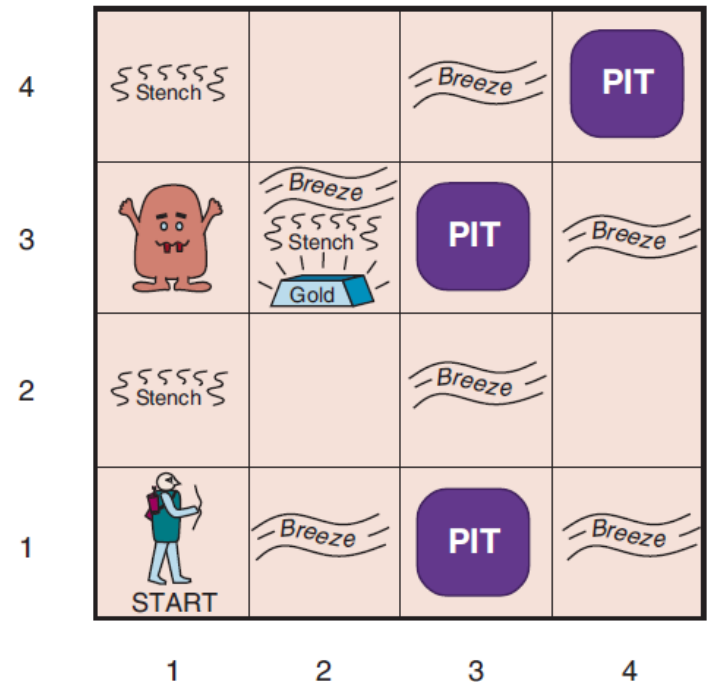
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$$KB = \{R_1, R_2, R_3, R_4, R_5\}$$



IV. Inference by Model Checking

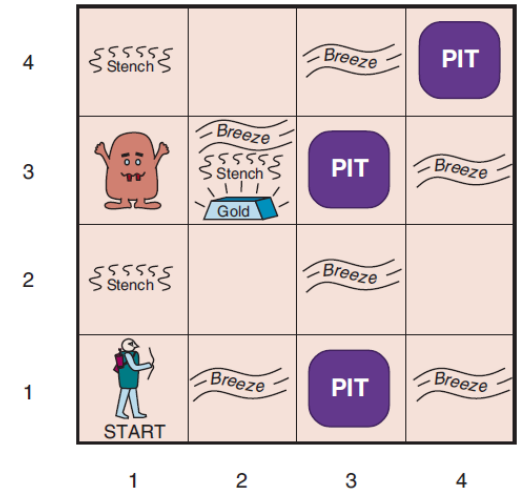
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Q. Does $KB \models \neg P_{1,2}$?



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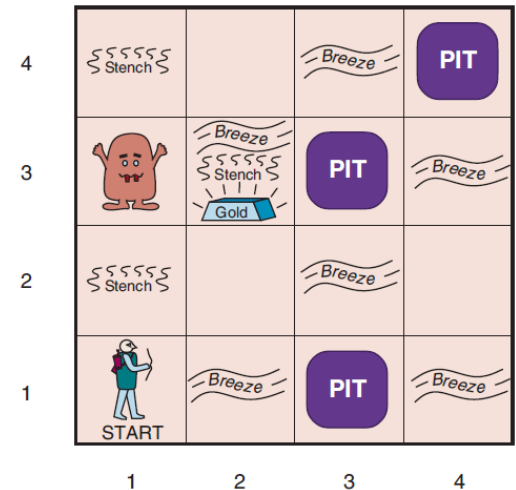
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Enumerate the models of KB and check if $\neg P_{1,2}$ is true in every model.

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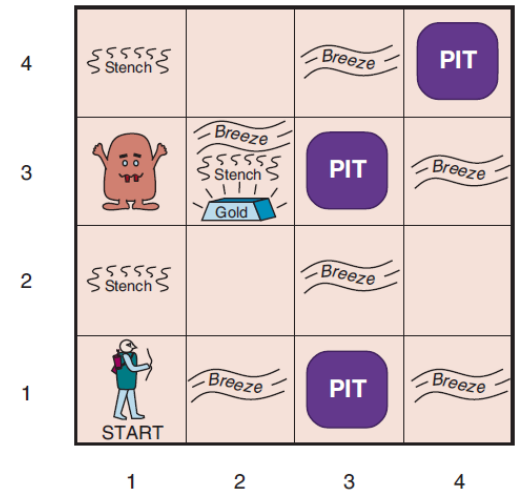
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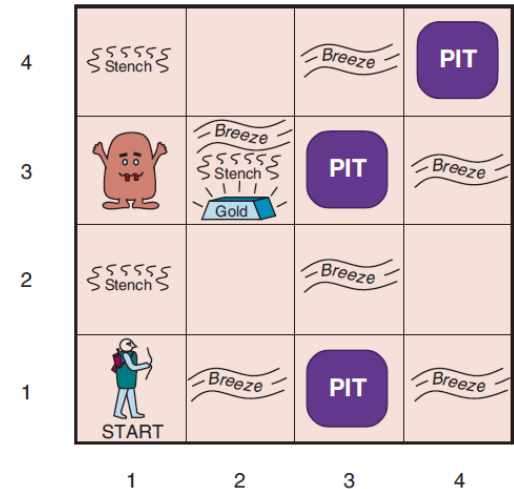
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$2^7 = 128$ possible models!

Truth Table Enumeration

128
rows

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>false</i>
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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<u><i>true</i></u>
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<u><i>true</i></u>
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<u><i>true</i></u>
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Truth Table Enumeration

$KB = \{R_1, R_2, R_3, R_4, R_5\}$ is true in only 3 models.

128
rows

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	false	false	false	false	false	false	true	true	true	true	false	false
false	false	false	false	false	false	true	true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
false	true	false	false	false	false	false	true	true	false	true	true	false
false	true	false	false	false	false	true	true	true	true	true	true	true
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false	true	false	false	false	true	true	true	true	true	true	true	true
false	true	false	false	true	false	false	true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	true	true	true	true	true	false	true	true	false	true	false

Truth Table Enumeration

$KB = \{R_1, R_2, R_3, R_4, R_5\}$ is true in only 3 models.

$\neg P_{1,2}$ is true in all 3.

128
rows

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	false	false	false	false	false	false	true	true	true	true	false	false
false	false	false	false	false	false	true	true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
false	true	false	false	false	false	false	true	true	false	true	true	false
false	true	false	false	false	false	true	true	true	true	true	true	true
false	true	false	false	false	true	false	true	true	true	true	true	true
false	true	false	false	false	true	true	true	true	true	true	true	true
false	true	false	false	true	false	false	true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	true	true	true	true	true	false	true	true	false	true	false

Truth Table Enumeration

$KB = \{R_1, R_2, R_3, R_4, R_5\}$ is true in only 3 models.
 $\neg P_{1,2}$ is true in all 3.
 $\Rightarrow KB \models \neg P_{1,2}$

128 rows

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	false	false	false	false	false	false	true	true	true	true	false	false
false	false	false	false	false	false	true	true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
false	true	false	false	false	false	false	true	true	false	true	true	false
false	true	false	false	false	false	true	true	true	true	true	true	true
false	true	false	false	false	true	false	true	true	true	true	true	true
false	true	false	false	false	true	true	true	true	true	true	true	true
false	true	false	false	true	false	false	true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	true	true	true	true	true	false	true	true	false	true	false

Truth Table Enumeration

$KB = \{R_1, R_2, R_3, R_4, R_5\}$ is true in only 3 models.
 $\neg P_{1,2}$ is true in all 3.
 $P_{2,2}$ is true in only 2 of 3.

$\Rightarrow KB \models \neg P_{1,2}$

128 rows

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	false	false	false	false	false	false	true	true	true	true	false	false
false	false	false	false	false	false	true	true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
false	true	false	false	false	false	false	true	true	false	true	true	false
false	true	false	false	false	false	true	true	true	true	true	true	true
false	true	false	false	false	true	false	true	true	true	true	true	true
false	true	false	false	false	true	true	true	true	true	true	true	true
false	true	false	false	true	false	false	true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	true	true	true	true	true	false	true	true	false	true	false

Truth Table Enumeration

$KB = \{R_1, R_2, R_3, R_4, R_5\}$ is true in only 3 models. $\Rightarrow KB \models \neg P_{1,2}$
 $\neg P_{1,2}$ is true in all 3.

$P_{2,2}$ is true in only 2 of 3. $\Rightarrow$ No inference of $KB \models P_{2,2}$

128
rows

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	false	false	false	false	false	false	true	true	true	true	false	false
false	false	false	false	false	false	true	true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
false	true	false	false	false	false	false	true	true	false	true	true	false
false	true	false	false	false	false	true	true	true	true	true	true	true
false	true	false	false	false	true	false	true	true	true	true	true	true
false	true	false	false	false	true	true	true	true	true	true	true	true
false	true	false	false	true	false	false	true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	true	true	true	true	true	false	true	true	false	true	false

Bad News

Suppose KB and α have n symbols. $\implies 2^n$ models!

The **propositional entailment problem** of showing $KB \models \alpha$ by truth table enumeration requires

$\Theta(2^n n)$ time

$O(n)$ space (not bad)

The problem is co NP-complete (likely not easier than NP-complete).

V. Logical Equivalence

Theorem proving: Apply rules of inference directly to the sentences in KB to construct a proof of a sentence without consulting models.

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$$(\alpha \equiv \beta)$$

$\alpha \equiv \beta$ if and only if $\alpha \models \beta$ and $\beta \models \alpha$.

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set of models for α

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$

$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$

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$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$

$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$

Validity

A sentence is *valid* if it is true in all models.

$$P \vee \neg P$$

Valid sentences are *tautologies*.

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$$P \vee \neg P$$

Valid sentences are *tautologies*.

Deduction theorem For any sentences α and β , $\alpha \models \beta$ if and only if the sentence $(\alpha \Rightarrow \beta)$ is valid.

We can decide if $\alpha \models \beta$ by checking that $\alpha \Rightarrow \beta$ is a tautology.

Satisfiability

A sentence is *satisfiable* if it is true in, or satisfied by, some model.

R_1	R_2	R_3	R_4	R_5	KB
<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
...					
<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<u><i>true</i></u>
<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<u><i>true</i></u>
<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<u><i>true</i></u>
<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>false</i>

$$R_1: \neg P_{1,1} \quad R_2: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

$$R_4: \neg B_{1,1} \quad R_3: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$

$$R_5: B_{2,1}$$

$(R_1 \wedge R_2 \wedge R_3 \wedge R_4 \wedge R_5)$ is satisfiable.

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<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
...					
<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<u><i>true</i></u>
<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<u><i>true</i></u>
<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<u><i>true</i></u>
<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
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The **SAT problem**: Determining the satisfiability of sentences in propositional logic.

Satisfiability

A sentence is *satisfiable* if it is true in, or satisfied by, some model.

R_1	R_2	R_3	R_4	R_5	KB
true	true	true	true	false	false
true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	false	true	true	false
...					
true	true	true	true	true	<u>true</u>
true	true	true	true	true	<u>true</u>
true	true	true	true	true	<u>true</u>
true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
false	true	true	false	true	false

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first NP-complete problem

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true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	false	true	true	false
...					
true	true	true	true	true	<u>true</u>
true	true	true	true	true	<u>true</u>
true	true	true	true	true	<u>true</u>
true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
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The **SAT problem**: Determining the satisfiability of sentences in propositional logic.

first NP-complete problem

$\alpha \models \beta$ if and only if the sentence $(\alpha \wedge \neg \beta)$ is unsatisfiable.

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R_1	R_2	R_3	R_4	R_5	KB
true	true	true	true	false	false
true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	false	true	true	false
...					
true	true	true	true	true	<u>true</u>
true	true	true	true	true	<u>true</u>
true	true	true	true	true	<u>true</u>
true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
false	true	true	false	true	false

$$\begin{aligned}
 R_1: & \neg P_{1,1} & R_2: & B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}) \\
 R_4: & \neg B_{1,1} & R_3: & B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1}) \\
 R_5: & B_{2,1}
 \end{aligned}$$

$(R_1 \wedge R_2 \wedge R_3 \wedge R_4 \wedge R_5)$ is satisfiable.

The **SAT problem**: Determining the satisfiability of sentences in propositional logic.

first NP-complete problem

Proof by contradiction

$\alpha \models \beta$ if and only if the sentence $(\alpha \wedge \neg \beta)$ is unsatisfiable.

Inference Rules

Modus Ponens

$$\frac{\alpha \Rightarrow \beta, \quad \alpha}{\beta}$$

$\alpha \Rightarrow \beta$
If **today is Tuesday**, then **John will go to campus**.
Today is Tuesday.
Therefore, **John will go to campus**.

Inference Rules

Modus Ponens

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If **today is Tuesday**, then **John will go to campus**.
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And-elimination

$$\frac{\alpha \wedge \beta}{\alpha}$$

$\alpha \wedge \beta$
A star is a sphere of gas, and **it is held together by its own gravity**.
A star is a sphere of gas.

Other Inference Rules

$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$ commutativity of $\wedge$

$(\alpha \vee \beta) \equiv (\beta \vee \alpha)$ commutativity of $\vee$

$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$ associativity of $\wedge$

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$\neg(\neg\alpha) \equiv \alpha$ double-negation elimination

$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$ contraposition

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$$\frac{\neg(\alpha \vee \beta)}{\quad}$$
$$\neg\alpha \wedge \neg\beta$$
$$\neg\beta$$

De Morgan

and-elimination

Applying Rules to the Wumpus World

KB:

$$\begin{aligned}
 R_1: \neg P_{1,1} \quad R_2: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}) \quad R_4: \neg B_{1,1} \\
 R_3: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1}) \quad R_5: B_{2,1}
 \end{aligned}$$

4	 Stench		 Breeze	 PIT
3		 Breeze  Stench  Gold	 PIT	 Breeze
2	 Stench		 Breeze	
1	 START	 Breeze	 PIT	 Breeze
	1	2	3	4

Applying Rules to the Wumpus World

4	 Stench		Breeze	PIT
3	 Stench	Breeze Stench Gold	PIT	Breeze
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 \end{aligned}$$

Proof for $\neg P_{1,2}$

Applying Rules to the Wumpus World

 Stench		 Breeze	 PIT
  Breeze  Stench  Gold		 PIT	 Breeze
 Stench		 Breeze	
 START		 Breeze	 PIT
1	2	3	4

KB:

$$\begin{aligned}
 R_1: \neg P_{1,1} \quad R_2: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}) \quad R_4: \neg B_{1,1} \\
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Applying Rules to the Wumpus World

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4	Stench	Breeze	PIT
3	Wumpus	Breeze Stench	PIT
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Searching for Proofs

- INITIAL STATE: the initial KB.
- ACTIONS: Apply any inference rule to any sentence that matches its top half.
- RESULT: Add the sentence in the bottom half of the inference rules.
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 - ♣ Search takes time $O(2^n n)$ – often not the worst case.

Monotonicity

The set of entailed sentences can only *increase* as new information is added to the KB.

if $KB \models \alpha$ then $KB \wedge \beta \models \alpha$

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An inference rule can be applied whenever its premise is found in the KB, regardless of what else is in the KB, and the conclusion must follow.